

ULTRA: Unified Multimodal Control for Autonomous Humanoid Whole-Body Loco-Manipulation

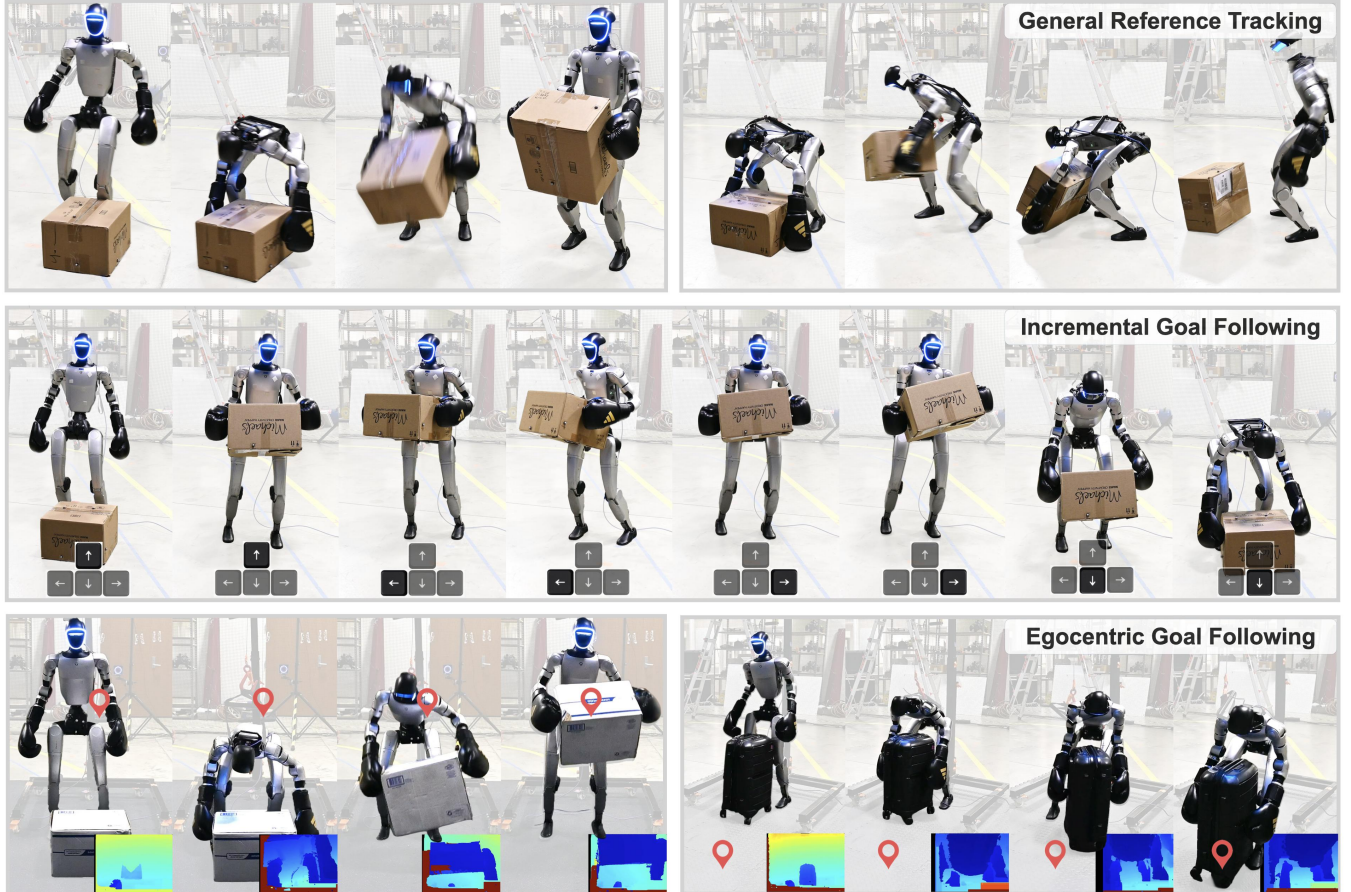


Fig. 1: ULTRA is an **all-in-one** controller for humanoid loco-manipulation that supports: **Top**, dense motion tracking from *arbitrary* reference. **Middle**, *fine-grained* control from operator commands. **Bottom**, *long-horizon* goal following with *egocentric perception*. It demonstrates autonomous whole-body behavior without relying on test-time motion references under real sensing. We provide additional examples in the supplementary video.

Achieving autonomous and versatile whole-body loco-manipulation remains a central barrier to making humanoids practically useful. Yet existing approaches are fundamentally constrained: retargeted data are often scarce or low-quality; methods struggle to scale to large skill repertoires; and, most importantly, they rely on tracking predefined motion references rather than generating behavior from perception and high-level task specifications. To address this limitations, we propose ULTRA, a unified framework with two key components. First, we introduce a physics-driven retargeting algorithm that translates large-scale motion capture to humanoid embodiments while preserving physical plausibility for contact-rich interactions. Second, we learn a unified multimodal controller that supports both dense references and sparse task specifications, under sensing ranging from accurate motion-capture state to noisy egocentric visual inputs. We distill a universal tracking policy into this controller, compress motor skills into a compact latent space, and apply reinforcement learning finetuning to expand

coverage and improve robustness under out-of-distribution scenarios. This enables coordinated whole-body behavior from sparse intent without test-time reference motions. We evaluate ULTRA in simulation and on a real Unitree G1 humanoid. Results show that ULTRA generalizes to autonomous, goal-conditioned whole-body loco-manipulation from egocentric perception, consistently outperforming tracking-only baselines with limited skills.

I. INTRODUCTION

Real-world loco-manipulation requires autonomy beyond replaying fixed reference motions. In unstructured environments, a humanoid must span a continuum: from dense motion references to sparse task goals, and from accurate state estimation to purely onboard sensing. Yet many controllers treat these as separate regimes and focus mainly on reference tracking [1], [2]. This fragmentation creates a

precision-flexibility trade-off: dense-tracking policies break down when references are missing or infeasible, while purely goal-conditioned policies often lack the fine-grained coordination needed for complex tasks. We therefore seek a unified controller that produces whole-body loco-manipulation and smoothly transitions between dense plans and sparse intent as information changes.

Despite progress in co-tracking humanoid and object dynamics [3], two bottlenecks hinder unified autonomy. First, kinematic retargeting can yield physically inconsistent demonstrations that fail in contact-rich tasks. Second, existing architectures typically assume a fixed conditioning structure tailored to one input type, and cannot interpret diverse or partial supervision within a consistent framework. Under shifting observability and goals at deployment, this rigidity leads to systemic instability. We address both barriers: limited, physically implausible demonstrations and policies designed mainly for tracking predefined trajectories rather than operating with subsets of conditioning signals.

To overcome the demonstration bottleneck, we introduce a physics-driven, *neural* retargeting algorithm that transfers large-scale motion capture (MoCap) to humanoid embodiments at scale. Unlike kinematic retargeting [2], [4], [5], which struggles to maintain physical consistency in contact-rich tasks, our retargeting is dynamics- and contact-aware by construction. We cast retargeting as simulation-constrained optimization with kinematic, dynamic, and contact constraints, and solve it with reinforcement learning (RL) at scale. Once trained, the policy generates large-scale physically feasible trajectories and generalizes to arbitrary data, enabling augmentation by scaling both objects and motions.

Building on this expanded corpus, we learn a Unified *muLT*imodal control for Autonomous humanoid control (ULTRA) that shifts from reference replay to perception-driven, goal-conditioned control. We first train a privileged universal tracker, then distill it into a student that follows diverse goal specifications, from dense references to sparse long-horizon targets (Fig. 1). This is enabled by (i) unified tokenization with availability masking [6], which keeps a single policy stable when references or modalities are missing; and (ii) a variational skill bottleneck plus RL finetuning [7] geared toward deployment with realistic perception and sensor noise. The bottleneck resolves ambiguity under sparse goals by maintaining coherent motion, while RL finetuning shifts control from reference-conditioned tracking to closed-loop goal stabilization under partial observability and distribution shift. Together, ULTRA yields one policy that tracks references when available and executes from egocentric perception and sparse intent when they are not.

In summary, ULTRA presents a unified system for practical whole-body loco-manipulation with three components: (i) a physics-driven neural retargeting pipeline that scales MoCap to humanoid embodiments and supports zero-shot augmentation; (ii) a versatile *multimodal* controller distilled from a privileged tracker that supports reference tracking and goal following across sensing modalities, including blind, MoCap-based, and depth-perception settings; and (iii) sim-

ulation and real-world evaluation on Unitree G1, showing a single unified model can outperform tracking-only baselines when references exist while enabling broader goal-conditioned behaviors as shown in Fig. 1.

II. RELATED WORK

A. Motion Retargeting

Retargeting transfers motion across embodiments with different morphologies. It originated in animation, where inverse-kinematics optimization adapted motions under kinematic constraints [8], and later evolved into learning-based mappings that amortized transfer for better generalization [9]. Humanoid retargeting requires stronger constraints because executability is contact-dependent and further limited by joint limits and dynamics. As a result, existing robot retargeting methods trade off efficiency and physical fidelity: kinematic approaches are fast but often under-model dynamics and degrade in contact-rich settings [10], [2], [4], [5], [6], while physics-based retargeting enforces contact and dynamics for physically plausible motions, but relies on non-convex, expensive optimization, typically per-trajectory RL [11] or costly sampling-based methods [12]. We target the missing regime: *physics-driven yet scalable* retargeting that preserves interaction semantics without per-trajectory RL. We perform dataset-scale retargeting with a single unified policy in one pass, and enable zero-shot augmentation to expand coverage.

B. Humanoid Whole-body Locomotion

Leveraging human motion data to teach humanoid robots complex skills has been widely studied. Early methods often use model-based control (e.g., trajectory optimization and MPC) to bridge embodiment and dynamics, while recent learning-based systems achieve precise tracking and agile motion replay [13], [14], [15], [16], [17], [1], [18], [19]. Beyond pure tracking, recent work moves toward foundation-style control by distilling large motion corpora into reusable priors, where a single model tracks diverse motions and supports multiple control modes [20], [21], [22], [23]. Others shape latent priors with adversarial RL [24], [25], [26], [27], but have not shown reliable scaling to large, heterogeneous loco-manipulation corpora. ULTRA follows the scalable teacher-student distillation paradigm but addresses a key bottleneck: offline distillation is limited by the state coverage of teacher rollouts. While less severe for humanoid-only control with more structured spaces, it becomes acute in high-dimensional robot-object interaction. To address this, we draw inspiration from animation practice [7] and use a two-stage pipeline: large-scale distillation followed by RL finetuning that *expands* interaction-state coverage and improves robustness to out-of-distribution goals and executions.

C. Humanoid Whole-body Loco-Manipulation

Most humanoid motion tracking emphasizes reproducing human motion on the robot and treats environmental dynamics as secondary [13], [28], [29], [30], which is brittle for contact-rich loco-manipulation. Recent work couples

humanoid motion and object interaction via co-tracking and shows strong agility [2], [1], [31], [32], but often assumes limited data replay or relies on external object state estimation (e.g., motion capture), limiting autonomy under onboard egocentric sensing. Other approaches use hierarchical designs that generate trajectories/keypoints and track them with a universal controller [33], [17]; however, stacking a high-level planner on a low-level controller can accumulate error and violate physical constraints. Adversarial motion priors broaden coverage but are typically task-specific, requiring careful objective engineering and scaling poorly to large, heterogeneous loco-manipulation corpora [34]. ULTRA addresses these issues by learning a goal-conditioned policy that unifies dense tracking and sparse task specifications in a *shared* latent space, and by using RL finetuning to induce *closed-loop* behaviors that expand interaction-state coverage. This yields a *versatile* single-policy controller under real-world perception and a *scalable* paradigm that leverages broad motion corpora.

III. PROBLEM FORMULATION AND PRELIMINARIES

A. Task Interface

We study whole-body loco-manipulation tasks where a humanoid interacts with a manipulated object, specified by a *goal* signal $c \in \mathcal{C}$ that defines the task objective. A rollout succeeds if the terminal outcome satisfies c , e.g., the humanoid root and/or the object reaches target transformations within a tolerance. At each time step t , the policy receives (i) an observation $\mathbf{o}_t \in \mathcal{O}$ and (ii) task conditioning $\mathbf{c}_t$, and outputs an action $\mathbf{a}_t \in \mathcal{A}$. Here $\mathbf{a}_t$ specifies target joint positions executed by a PD controller.

Goal specification. We consider two forms of c : (i) *dense reference conditioning*, which provides a time-indexed motion reference and thus specifies intermediate motions; and (ii) *sparse goal conditioning*, which specifies long-horizon target transformations for the humanoid root and/or object while leaving intermediate motions underdetermined.

Perception. Beyond proprioception, we consider two regimes for object sensing: (i) *MoCap-based sensing*, where $\mathbf{o}_t$ includes accurate object pose (e.g., from motion capture); and (ii) *egocentric depth perception*, where $\mathbf{o}_t$ includes an egocentric point cloud from a depth sensor (e.g., head-mounted), from which object state must be inferred.

B. Preliminaries

Since the controller may rely on partial onboard sensing, we model loco-manipulation as a goal-conditioned *Partially Observable Markov Decision Process* (POMDP). Let $\mathbf{s}_t \in \mathcal{S}$ be the underlying system state (humanoid and scene, including the object), with dynamics $\mathbf{s}_{t+1} \sim \mathcal{T}(\mathbf{s}_t, \mathbf{a}_t)$. The policy acts from $\mathbf{o}_t = \Omega(\mathbf{s}_t)$ and conditioning $\mathbf{c}_t$, producing $\mathbf{a}_t \in \mathcal{A}$. We optimize $\pi(\mathbf{a}_t | \mathbf{o}_t, \mathbf{c}_t)$ to maximize expected discounted return: $\max_{\pi} \mathbb{E}[\sum_{t \geq 0} \gamma^t r(\mathbf{s}_t, \mathbf{a}_t, \mathbf{c}_t)]$. The following sections describe how we use PPO [35] and imitation to learn policies, including the observation/reward design and key techniques for our tasks.

IV. METHOD

As shown in Fig. 2, ULTRA follows a four-stage training paradigm that couples physics-driven motion retargeting with teacher-student learning. In Stage 1, we learn a *retargeting policy* that maps human MoCap motions to physically feasible humanoid loco-manipulation rollouts. In Stage 2, we train a privileged *teacher policy*, leveraging full state and dense reference trajectories from the retargeted rollouts. In Stage 3, we distill the teacher into a *multimodal student* that operates under perception and sparse goal specifications. Finally, we deploy the student with separated control mode.

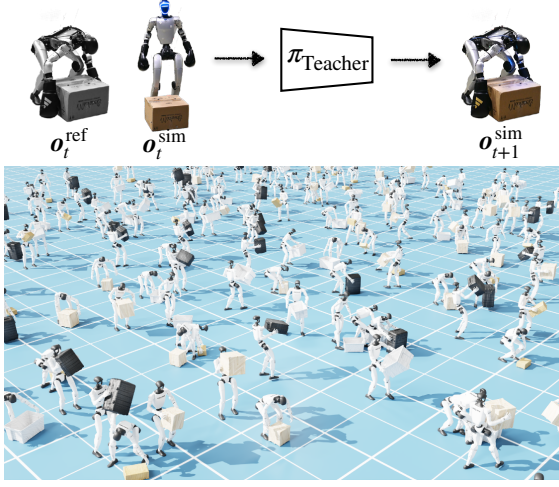
A. General Motion Tracking for Neural Retargeting

Given a human-object demonstration represented by an SMPL-X [36] motion sequence and an object pose trajectory, our goal is to generate a physically feasible rollout on the target humanoid (e.g., Unitree G1) that preserves the overall motion and intended interaction. Traditional retargeting solves inverse kinematics under kinematic constraints. We instead cast retargeting as RL-based trajectory optimization: rewards encode tracking, while simulator transitions enforce kinematics, dynamics, and contacts. Following [3], this is well suited for contact-rich loco-manipulation, where contacts are hard to express as kinematic constraints. As preprocessing, we scale the human-object trajectory to match G1 and define a fixed correspondence from human key links to humanoid counterparts. We then train a *unified* retargeting policy across all motions, producing physically consistent rollouts without per-motion optimization or re-training. Dense, full-body tracking is brittle under embodiment mismatch and becomes especially fragile during object interaction, where exact link-wise targets may be infeasible and contact often requires deliberate deviations. Our key insight is to combine (i) relaxed tracking that prioritizes end effectors critical for loco-manipulation with (ii) interaction and contact rewards that correct mismatch-induced errors.

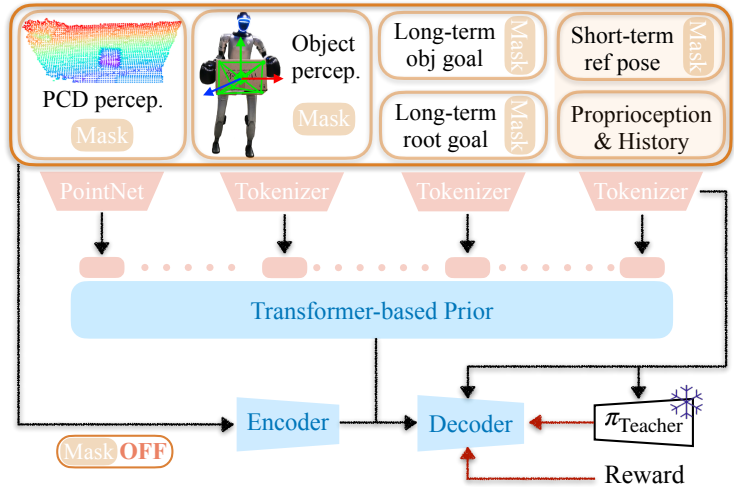
Reward. We define $r_{\text{track}} = r_p \cdot r_r \cdot r_{\text{obj}} \cdot r_{\text{int}} \cdot r_{\text{ct}} \cdot r_{\text{eng}}$, with all terms computed in a heading-aligned humanoid frame. Let $\mathcal{F}$ include only feet and palms. r_p tracks end-effector positions as sparse anchors; r_r matches normalized link directions over a fixed key edge set; and r_{eng} regularizes joint effort and foot placement. To reduce ambiguity, r_{obj} tracks object pose/velocities and r_{int} matches palm-to-surface offsets over sampled object points. We also align contact events by mapping contacts on human links to corresponding humanoid links, yielding r_{ct} .

Observation. The policy uses a privileged, reference-aware observation containing simulator state and its deviation from the SMPL-X reference. Since preprocessing establishes a fixed correspondence after scaling/alignment, residuals are well-defined: $\mathbf{o}_t = [\mathbf{o}_t^{\text{sim}}, \mathbf{o}_t^{\text{ref}}, \mathbf{o}_t^{\Delta}]$. $\mathbf{o}_t^{\text{sim}}$ includes proprioception and contact signals; $\mathbf{o}_t^{\text{ref}}$ provides selected correspondence-defined reference quantities (including object state); and $\mathbf{o}_t^{\Delta}$ encodes heading-aligned simulation-reference differences. All quantities are expressed in a heading-aligned frame to remove global yaw.

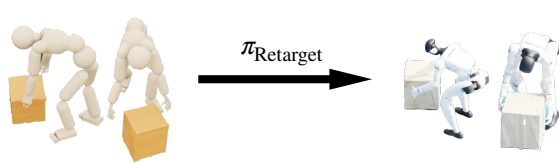
Stage 1: Teacher - General Motion Tracking



Stage 2: Student - Multimodal Learning



Stage 0: Tracking for Motion Retargeting



Stage 3: Student - Deployment

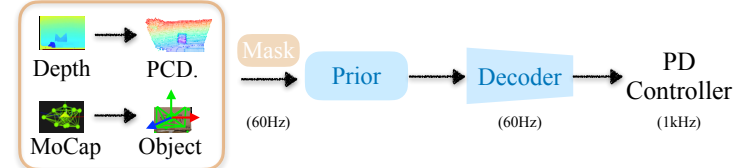


Fig. 2: ULTRA follows four stages: (i) **Neural Retargeting**: an RL policy converts MoCap data into physically feasible G1 rollouts with augmentation; (ii) **Tracking**: a privileged teacher tracks these rollouts using full state and references; (iii) **Distillation**: we distill the teacher into a multimodal student for realistic sensing and sparse goals, with additional RL finetuning; (iv) **Deployment**: the student runs under real sensing, supporting depth input or MoCap-based state estimation.

State initialization and early termination. Because we cannot reliably initialize the humanoid from an SMPL-X pose, we do not use reference-state initialization [37]. Each episode starts from a default standing pose, initially tracking the first reference frame to stabilize the humanoid before transitioning to full tracking with smoothly varying weights. We terminate on falls, excessive deviation, or contact mismatch for 20 frames [3] to improve sample efficiency.

Simplified actuation. Since retargeting is used only to *generate reference rollouts*, we prioritize motion quality and throughput over hardware-faithful control. We use an *idealized* low-level controller in simulation (i.e., control frequency equal to simulation frequency), enabling stronger, more responsive tracking than onboard PD control. We train without domain randomization or perturbations, and address robustness later in Sec. IV-B.

Trajectory and object augmentation. RL-based retargeting also enables *flexible augmentation* (Fig. 4). Since preprocessing already scales positions, we can (i) apply anisotropic scaling along coordinate axes and (ii) scale the manipulated object with different coefficients, while interaction/contact rewards correct imperfections and the simulator enforces physical feasibility. Crucially, these augmentations are handled by a *single retargeting policy without retraining*.

B. Dense Motion Tracking for Teacher Policy

Sec. IV-A converts human-object demonstrations into physically feasible G1 rollouts. For downstream imitation, we train a separate privileged teacher π_{teacher} to track these

rollouts. The teacher uses the deployment control interface and actuation limits, but trains with privileged state and dense reference residuals to accelerate learning. We randomize physics and inject perturbations to broaden state visitation and teach recovery, producing stable behaviors that provide high-quality supervision for the student.

Observation. The teacher uses the same reference-aware observation as retargeting, but does not require cross-embodiment correspondence since the reference is already in the humanoid embodiment.

Dense tracking objective. The teacher uses the same reward template, but replaces sparse anchoring with *full* link tracking, together with object, interaction, and contact reward.

Reference initialization and robustness training. We initialize from randomly sampled reference frames and include occasional stand still episodes that track standing references, reflecting deployment from a stable standing pose. To improve robustness, we randomize humanoid/object physical properties and inject perturbations, with a short grace period to allow recovery. We use the same early-termination criteria as retargeting and add no observation noise at this stage.

C. Multimodal Student Policy

We distill the privileged teacher into a multimodal student policy π_{student} . Unlike the teacher, the student observes only partial state and conditions on whatever modalities are available at test time via an availability mask randomly sampled during training. This retains teacher behavior as a prior while enabling goal-reaching under missing observations.

Multimodal observation with availability mask. The student consumes heterogeneous inputs: $\mathbf{o}_t^{\text{student}} = [\mathbf{o}_t^{\text{proprio}}, \mathbf{o}_t^{\text{goal}}, \mathbf{o}_t^{\text{object}}, \mathbf{o}_t^{\text{pcd}}, \mathbf{m}_t]$. $\mathbf{o}_t^{\text{proprio}}$ contains proprioception (e.g., joint states, IMU), $\mathbf{o}_t^{\text{object}}$ provides object state (e.g., MoCap), and $\mathbf{o}_t^{\text{pcd}}$ is an egocentric point cloud (e.g., egocentric camera). $\mathbf{o}_t^{\text{goal}}$ encodes task objectives and commands, including (i) long-horizon object transforms, (ii) long-horizon humanoid root transforms, and (iii) next-frame humanoid local state changes for tracking. We also include discretized commands (e.g., stand still) for deployment. $\mathbf{m}_t$ indicates which modalities are present.

Distillation. We collect data with a DAgger-style loop [38]: we roll out with the teacher initially, gradually shift to the student, and query the teacher on visited states to obtain $\mathbf{a}_t^{\text{teacher}}$. During training, an encoder $q_\phi(z_t^{\text{res}} | \mathbf{o}_t^{\text{student}}, \mathbf{o}_t^{\text{teacher}})$ infers a latent residual [6] using privileged teacher inputs, while a prior $p_\theta(z_t^{\text{prior}} | m(\mathbf{o}_t^{\text{student}}))$ predicts a latent from masked student observations ($m(\cdot)$ applies $\mathbf{m}_t$). We combine them as $z_t = z_t^{\text{prior}} + z_t^{\text{res}}$ and sample actions $\mathbf{a}_t^{\text{student}} \sim \pi_{\text{student}}(\mathbf{a}_t | \mathbf{o}_t^{\text{student}}, z_t)$. We implement π_{student} with a transformer-based encoder [39] that projects each modality into shared tokens; $\mathbf{m}_t$ gates tokens and modulates cross-modal attention to ignore missing inputs. At deployment, we sample z_t from the prior only.

Training objective. We match teacher actions while aligning the prior with the privileged posterior:

$$\mathcal{L} = \|\mathbf{a}_t^{\text{student}} - \mathbf{a}_t^{\text{teacher}}\|_2^2 + \mathcal{L}_{\text{aux}} + \lambda_{\text{KL}} D_{\text{KL}}(q_\phi(z_t | \mathbf{o}_t^{\text{student}}, \mathbf{o}_t^{\text{teacher}}) \| p_\theta(z_t | \mathbf{o}_t^{\text{student}})). \quad (1)$$

$\mathcal{L}_{\text{aux}}$ uses reconstruction heads (recovering masked modalities) to encourage z_t to retain task-relevant information.

Curriculum learning. Beyond DAgger, we use two curricula to keep the prior effective under partial observability: we progressively increase modality-masking probability, and anneal λ_{KL} and auxiliary weights to avoid posterior collapse while preserving latent skill diversity.

Shortcut for tracking. For local-goal tracking, behavior is largely deterministic, so a stochastic latent helps less. We add a residual shortcut (with the mask) from the full-body goal directly to the decoder, preserving low-level reference information and stabilizing decoding (Fig. 2).

RL finetuning. We perform RL finetuning on top of the distilled student by switching a subset of parallel environments to a goal-reaching objective while continuing distillation updates. Following [7], we partition simulators into (i) distillation environments replaying reference motions with imitation losses, and (ii) RL environments optimizing task success under state/goal perturbations. We sample random offsets for the object goal, humanoid root goal, and their initializations.

Deployment versatility. At test time, the student receives only $\mathbf{o}_t^{\text{student}}$ and samples z_t from the prior. With the same parameters, modality masking enables: (i) high-fidelity tracking by unmasking local reference (Fig. 1 Top), (ii) goal-conditioned control by masking local reference and unmasking long-horizon goals (Fig. 1 Middle), and (iii) vision-

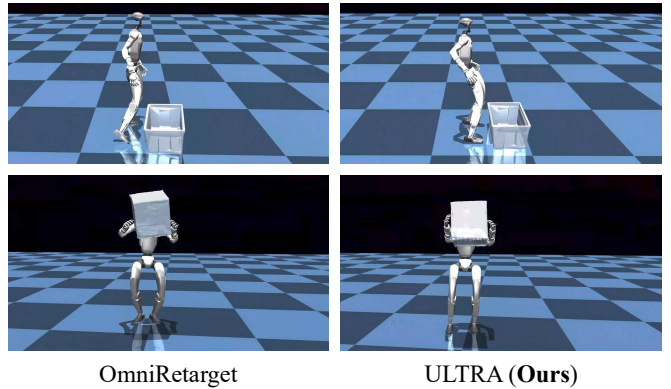


Fig. 3: Qualitative comparison of our retargeting and OmniRetarget [2] at the same frame/sequence. **Top:** final frame; the baseline shows undesired standing foot placement. **Bottom:** a contact frame; ours yields more stable contacts.

based manipulation by masking MoCap object state while unmasking point clouds (Fig. 1 Bottom).

V. EXPERIMENTAL RESULTS

We evaluate ULTRA end-to-end for autonomous whole-body loco-manipulation, from data generation to real-world transfer. We ask: (i) Can we retarget human-object MoCap into physically consistent rollouts with stable contacts and minimal sliding/penetration? (ii) Under dense references, can the student match a privileged teacher and specialized trackers? (iii) Under sparse goals, does RL finetuning improve robustness and yield a semantically organized latent skill space? (iv) Can one policy transfer to a real humanoid without test-time references? We evaluate four axes: retargeting, tracking, goal execution, and real deployment on Unitree G1.

A. Experimental Setup

Simulation. We train in IsaacGym [40] with GPU-parallel environments and validate key results in MuJoCo [41]. Real trials use a physical Unitree G1 [42].

Dataset. We use OMOMO [43] human-object MoCap, using the corrected subset from [3] for a fair comparison with [2]. We focus on 4 box-shaped objects (others require dexterous hands). We retarget all sequences with our RL-based pipeline (Sec. IV-A) and augment via anisotropic trajectory scaling and object resizing, yielding a $\sim 6\times$ larger corpus (Fig. 4). We use the same train/test split for ID evaluation and define OOD by held-out motions and novel object scales.

B. Motion Retargeting

Baselines. We compare against: (i) PHC [4] (kinematics-based retargeting for G1), (ii) GMR [5] (humanoid motion retargeting without objects), and (iii) OmniRetarget [2] (interaction-preserving kinematic engine with interaction-mesh augmentation). All retarget from the same OMOMO subset processed by [3].

Metrics. We measure: (i) *penetration* (duration, max depth) between humanoid/object/environment [2]; (ii) *foot skating* (sliding duration, max tangential stance velocity), with stance defined geometrically (foot within 2 cm of ground) to avoid

TABLE I: Motion-tracking evaluation in IsaacGym. All methods are trained/evaluated on our data unless noted. **Green** highlights our primary tracking controller.

Method	In-Distribution (ID)							Out-of-Distribution (OOD)						
	Succ $\uparrow$	Succ $\uparrow$	Humanoid			Object		Succ $\uparrow$	Succ $\uparrow$	Humanoid			Object	
	(Humanoid)	(+Object)	$E_{g\text{-mpipe}} \downarrow$	$E_{\text{mpipe}} \downarrow$	$E_{\text{jitter}} \downarrow$	$E_{\text{pos}} \downarrow$	$E_{\text{rot}} \downarrow$	(Humanoid)	(+Object)	$E_{g\text{-mpipe}} \downarrow$	$E_{\text{mpipe}} \downarrow$	$E_{\text{jitter}} \downarrow$	$E_{\text{pos}} \downarrow$	$E_{\text{rot}} \downarrow$
(a) Unified Multimodal Controller														
ULTRA (Ours)	67.30 $\pm$ 0.12	57.44 $\pm$ 0.40	13.49 $\pm$ 0.14	5.89 $\pm$ 0.02	6.27 $\pm$ 0.00	53.42 $\pm$ 0.21	65.44 $\pm$ 0.37	70.57 $\pm$ 0.54	52.00 $\pm$ 0.44	35.55 $\pm$ 0.23	14.67 $\pm$ 0.08	6.81 $\pm$ 0.01	56.52 $\pm$ 0.35	67.60 $\pm$ 0.58
(b) Privileged Teacher														
ULTRA Teacher	97.57 $\pm$ 0.05	89.79 $\pm$ 0.11	12.98 $\pm$ 0.30	5.64 $\pm$ 0.05	14.81 $\pm$ 0.08	17.15 $\pm$ 0.03	23.28 $\pm$ 0.33	97.12 $\pm$ 0.43	81.33 $\pm$ 0.78	19.14 $\pm$ 0.48	7.94 $\pm$ 0.11	15.91 $\pm$ 0.08	25.57 $\pm$ 0.28	33.49 $\pm$ 0.37
(c) General Motion Tracking														
ULTRA (RL)	54.47 $\pm$ 0.43	41.78 $\pm$ 0.31	49.30 $\pm$ 0.32	16.23 $\pm$ 0.11	20.04 $\pm$ 0.15	47.48 $\pm$ 0.31	60.53 $\pm$ 0.09	53.38 $\pm$ 0.98	23.54 $\pm$ 0.24	68.11 $\pm$ 0.26	22.22 $\pm$ 0.01	17.46 $\pm$ 0.09	66.17 $\pm$ 0.54	59.44 $\pm$ 0.73
ULTRA (Distillation)	85.03$\pm$3.00	77.15$\pm$0.57	15.45$\pm$0.08	6.84$\pm$0.04	8.12$\pm$0.01	25.48$\pm$0.48	33.97$\pm$0.58	86.63$\pm$0.50	52.74$\pm$0.04	35.01$\pm$0.31	13.48$\pm$0.10	9.35$\pm$0.01	36.18$\pm$0.30	38.18$\pm$0.29
HDMI [1]	13.07 $\pm$ 0.20	9.94 $\pm$ 0.38	92.77 $\pm$ 0.56	26.90 $\pm$ 0.10	26.13 $\pm$ 0.60	78.93 $\pm$ 0.42	70.23 $\pm$ 0.62	13.92 $\pm$ 0.78	12.95 $\pm$ 0.30	87.07 $\pm$ 0.44	27.54 $\pm$ 0.06	29.19 $\pm$ 0.38	77.33 $\pm$ 2.27	71.16 $\pm$ 0.48
OmniRetarget [†] [2]	41.27 $\pm$ 1.17	21.90 $\pm$ 0.29	62.96 $\pm$ 1.43	15.37 $\pm$ 0.17	39.35 $\pm$ 0.57	77.94 $\pm$ 3.52	66.47 $\pm$ 1.15	33.36 $\pm$ 0.39	20.78 $\pm$ 0.13	74.80 $\pm$ 0.34	16.23 $\pm$ 0.15	49.52 $\pm$ 0.52	55.11 $\pm$ 2.32	62.44 $\pm$ 0.77
OmniRetarget [2]	51.34 $\pm$ 0.67	20.91 $\pm$ 0.52	67.12 $\pm$ 0.86	7.43 $\pm$ 0.07	39.92 $\pm$ 1.44	60.67 $\pm$ 0.54	67.03 $\pm$ 0.19	46.71 $\pm$ 0.74	25.82 $\pm$ 0.52	68.34 $\pm$ 0.82	8.98 $\pm$ 0.19	40.08 $\pm$ 1.77	58.57 $\pm$ 2.37	66.70 $\pm$ 0.84

[†] Trained/evaluated on original OmniRetarget dataset.

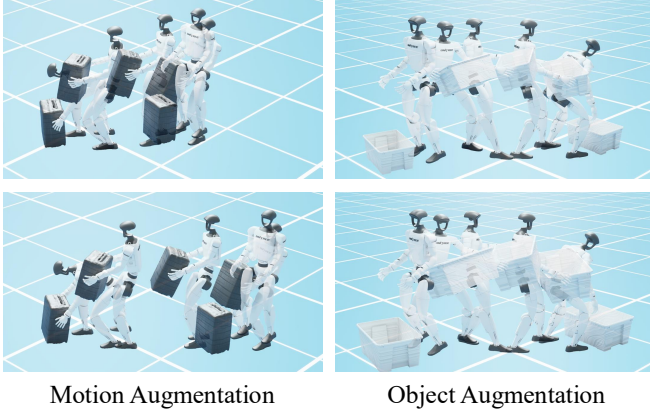


Fig. 4: Zero-shot augmentation with the retargeting policy. **Left:** trajectory scaling. **Right:** object scaling. Motions remain plausible, enabling scalable data augmentation.

noisy MoCap stance labels; and (iii) *contact floating*, the duration of lost hand-object contact during transport, detected via MuJoCo contact queries.

Quantitative evaluation. Table II shows ULTRA outperforms baselines across nearly all metrics/categories: lowest foot-skating duration/velocity and much less contact floating (near-zero on Largebox/Suitcase), while also reducing penetration. We attribute this to physics-aware retargeting that enforces contact/dynamics, keeping stance feet planted and preserving hand-object contact when lifting.

Qualitative evaluation. Fig. 3 shows more accurate hand/foot placement than OmniRetarget, whose kinematic formulation often breaks contact consistency and yields unnatural configurations relative to the object and ground.

Effectiveness of data augmentation. Our augmentation diversifies motions without retraining the retargeter and applies along the full trajectory (not only the initial frame), producing temporally consistent variations (Fig. 4). This improves downstream generalization: in Table I, OmniRetarget retrained on our augmented data attains substantially higher OOD tracking success than when trained on its original dataset, confirming broader state/skill coverage.

C. General Motion Tracking

Baselines. We evaluate dense tracking (full reference provided) against: (i) OmniRetarget[†] (original data), (ii)

OmniRetarget retrained on our augmented set, and (iii) HDMI [1] adapted to our setting. We also report ULTRA ablations: (i) direct RL under student observations (tracking only), (ii) tracking-only distillation, and (iii) all-task unified training. The privileged teacher is an upper bound.

Metrics. We report success (Succ): no fall and per-frame $E_{g\text{-mpipe}} < 0.3\text{m}$ and $E_{\text{pos}} < 0.3\text{m}$; we also report humanoid-only success. Tracking errors include $E_{g\text{-mpipe}}$, E_{mpipe} , E_{jitter} , and object errors E_{pos} , E_{rot} .

Results. Table I shows ULTRA strongly outperforms baselines for humanoid-object tracking, especially under OOD motions/object scales. HDMI often becomes unstable at our scale and fails to converge. OmniRetarget trains smoothly but frequently fails manipulation: humanoid success is reasonable, but drops when object tracking is required, likely due to missing explicit object observations and a default-to-locomotion failure mode. ULTRA closes this gap via a privileged teacher with object/contact signals and distillation that preserves closed-loop tracking under partial observability.

Distillation vs. direct RL. Direct RL under partial observations must learn stabilization and sustained contact jointly, so early failures dominate and training can collapse. Distillation transfers contact-aware corrections learned with privileged state and dense references, yielding higher success.

Distillation regularizes control. Despite less information, the student attains lower jitter than the teacher, suggesting imitation suppresses high-frequency RL micro-corrections that reduce instantaneous error but induce chatter.

All-task training induces a prior. Unified training (ours) slightly reduces ID tracking success versus *tracking only* but largely preserves OOD performance, suggesting learning contact-stable primitives rather than memorization.

D. Goal-Conditioned Following

Metric. Success (Succ): no fall and terminal state within 0.3m of the goal.

Comparisons. Tracking-only baselines (e.g., HDMI [1], OmniRetarget [2]) require dense references and are inapplicable; we compare ULTRA to ablations.

Tasks. We test two perception regimes: (i) *MoCap* object pose and (ii) *egocentric vision* from a noisy depth point cloud. In both, we provide only long-horizon object target.

TABLE II: Physical interaction quality in simulation. Lower is better.

Method	Penetration		Foot Skating		Contact Floating
	Duration ↓	Max Depth (cm) ↓	Duration ↓	Max Vel. (cm/s) ↓	Duration ↓
Largebox					
PHC [4]	0.908 ± 0.125	0.073 ± 0.048	0.303 ± 0.145	0.032 ± 0.022	0.025 ± 0.054
GMR [5]	0.522 ± 0.259	0.086 ± 0.053	0.366 ± 0.317	0.029 ± 0.020	0.111 ± 0.171
OmniRetarget [2]	0.000 ± 0.002	0.013 ± 0.002	0.205 ± 0.106	0.035 ± 0.019	0.231 ± 0.224
ULTRA (Ours)	0.008 ± 0.030	0.012 ± 0.002	0.061 ± 0.031	0.018 ± 0.010	0.015 ± 0.063
Suitcase					
PHC [4]	0.914 ± 0.119	0.077 ± 0.051	0.286 ± 0.147	0.035 ± 0.024	0.032 ± 0.065
GMR [5]	0.571 ± 0.265	0.105 ± 0.050	0.399 ± 0.368	0.028 ± 0.018	0.142 ± 0.175
OmniRetarget [2]	0.003 ± 0.016	0.012 ± 0.002	0.264 ± 0.141	0.040 ± 0.021	0.404 ± 0.279
ULTRA (Ours)	0.002 ± 0.013	0.017 ± 0.019	0.062 ± 0.045	0.017 ± 0.008	0.008 ± 0.040

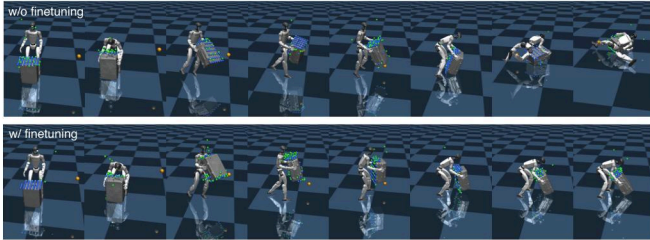


Fig. 6: Sim-to-sim comparison for egocentric goal following. Blue/green: point cloud without/with noise; yellow: object goal. **Top:** without finetuning. **Bottom:** with finetuning.

RL finetuning expands OOD coverage. Table III and Fig. 6 show finetuning yields modest ID gains but large OOD gains under random goal offsets (nearly doubling under point clouds and tripling under position-only). This suggests finetuning expands interaction-state coverage and reinforces closed-loop recovery beyond the demonstration manifold.

Latent structure. Fig. 5 shows the latent separates tracking vs. goal following across modalities while remaining on a shared manifold. Tracking stays distinct because we bypass the stochastic latent via a residual shortcut when local tracking goals are provided (Sec. IV-C), leaving the latent to capture multimodality under sparse goals. The latent also aligns with text-derived semantic clusters (MiniLM [44] + K-Means), indicating organization by control regime and intent.

E. Real-World Deployment

Tasks. We deploy ULTRA on a physical Unitree G1. The student runs onboard at the control frequency with proprioception and, when available, OptiTrack object pose. For dense tracking, we test OMOMO subsets (bimanual box lift/carry, suitcase transport) with household objects. For goal-conditioned control, we provide no motion references and specify future object transforms via keyboard commands.

Point cloud extraction. For egocentric perception, we extract object point clouds from depth only: back-project depth pixels using calibrated intrinsics, crop a forward ROI, remove the ground plane, take the dominant cluster as the box, and downsample to a fixed size for policy input.

Quantitative evaluation. Table IV reports success rates: the policy reliably grasps/transport on hardware and achieves reasonable sparse-goal success under out-of-distribution operator commands, including composed motions.

Failure analysis. Failures mainly arise from (i) friction gaps causing occasional grasp slip, (ii) depth noise/occlusion

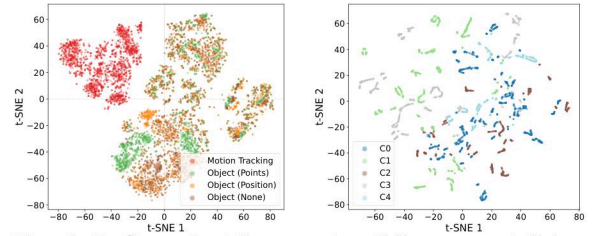


Fig. 5: **Left:** embeddings under different modalities; aside from tracking, embeddings largely mix, indicating a shared latent space. **Right:** latents cluster by text labels (C0–C4), showing semantic structure.

TABLE III: Sim-to-sim success rate on Mujoco across goal type with in-distributional (ID) goals from training and out-of-distributional (OOD) goals with random offsets, and across perception with egocentric point clouds or object position with no shape. Policies are trained in IsaacGym and evaluated in MuJoCo with 20 selected motion per setting.

RL finetuning	ID Goals		OOD Goals	
	Points	Position	Points	Position
✗	16 / 20	14 / 20	5 / 20	4 / 20
✓	19 / 20	16 / 20	9 / 20	12 / 20
Δ (RL gain)	+18.8%	+14.3%	+80.0%	+200.0%

TABLE IV: Real-world success rates on the OMOMO subset using a Unitree G1 humanoid. Each task is evaluated over two trials. MoCap provides object pose tracking for non-egocentric control modes, while the egocentric setting relies only on onboard sensing. MoCap is used for success evaluation in all settings. Dense reference tracking is direction-agnostic and thus reported as a single success rate.

Setting	Vertical	Lateral
Dense Reference Tracking	73% (19/26)	
Sparse Goal Following (MoCap)	80% (8/10)	90% (9/10)
Sparse Goal Following (Egocentric)	50% (5/10)	60% (6/10)

breaking point-cloud extraction, and (iii) disturbances beyond the recovery margin learned with domain randomization, motivating future tactile integration.

VI. CONCLUSION

ULTRA is a unified framework for practical humanoid whole-body loco-manipulation that moves beyond reference replay toward perception- and goal-driven autonomy. It combines an RL-formulated, physics-driven retargeting policy that scales human-object MoCap into physically consistent humanoid rollouts with a distilled multimodal controller that unifies dense tracking and sparse goal specification. Experiments show improved interaction fidelity from retargeting, a student that matches tracking performance while remaining robust under distribution shift, and RL finetuning that boosts success on out-of-distribution goals. We further validate sim-to-real transfer on Unitree G1, demonstrating reliable dense tracking and sparse goal following. Overall, ULTRA points to a scalable path for versatile loco-manipulation that adapts online from realistic sensing without test-time references.

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